MILITARY SPECIFICATION

COMPRESSORS, RECIPROCATING, MOTOR DRIVEN,

NAVAL SHIPBOARD HIGH PRESSURE AIR SERVICE

This amendment forms a part of Military Specification MIL-C-18419B(SHIPS). 29 June 1962

Pages 1 and 2, paragraph 2.1: Make the following changes:

Under "Specifications, Federal"' add:

" F F - B - 1 8 5 Bearings, Roller. Cylindrical: and Bearings. Roller. Self-aligning. "

Under "Standards. Military" add:

"MIL-STD-276 - Impregnation of Porous Nonferrous Metal Castings
"MIL-STD-438 - Schedule of Piping, Valves, Fittings and Associated Piping Components for Submarine Service
"MIL-STD-777 - Schedule of Piping, Valves, Fittings and Associated Piping Components for Surface Ships "'

Under "Drawings, Bureau of Ships" delete "'5000 -S4800-3000" and add

"810-1385941 - Fittings, Sil Brzg, WOG, 3000 psi for U T Inspection
"810-1385942 - Fittings, Bronze Sil Brzg . WOG, [or U T Inspection
"'810-1385943 - Unions, Sil Brzg . 3000 psi WOG IPS for U T Inspection
"810-1385944 - Fittings, Sil Brzg OD 3000 *psi* WOG FORU T Inspection
810-1385945 - Bushings. Re ducing CU-NI Sil Brzg WOG for U T Inspection
"810-1385946 - Unions Bronze Sil Brzg, WOG for U T inspection
'"810-1385948 - Unions Sil Brzg , OD 3000 PSI, WOG for U T Inspection
"810-1385950 - Bosses. Non-Ferrous Piping for U T Inspection
"810-1385951 - Bushings, Reducing CU-NI Sil Brzg , IPS, 3000 psi, WOG for U T Inspection
"810-1325952 - Bushings, Reducing CU-NI Sil **Br** zg **,** OD 3000 psi, WOG for U T inspection

Under "Publications Bureau of Ships' delete "NAVSHIPS 250-637-2" and "NAVSHIPS 250-548-8" and add

NAVSHIPS 250-634-6 - Fabrication and Inspection of Brazed Piping Systems

Page 2, paragraph 2.2: Under "'Compressed Air and Gas Institute" m third line change "2nd edition" t "3rd edition"

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1 CHANGES FROM PREVIOUS ISSUE THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED"#"' TO IN DICATE WHERE CHANGES DELETIONS A DDITIONS. ETC , FROM THE PREV OUS ISSUE HAVE BEEN MADE TH IS HAS BE EN DONE AS A CONV E NIENCE ONL Y AND THE GO ERNM ENT A SSUM ES NO LIABI LITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIO BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUM ENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE

FSC 43

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Page 3, paragraphs 3.3 through 3.3 1.3: Delete and substitute:

"3.3 General design, -

"3 .3.1 The principle of maximum reliability is paramount and no compromise of this principle shall be made with any other basic requirements of design, Where wear or corrosion is unavoidable, the parts subjected to such wear or corrosion shall be of the best material for the purpose in order to reduce those detrimental effects to a minimum. It is the intention of this specification to obtain air compressors of such design that they will have an operating life of at least 50,000 hours with no li mitation on the number of starts and stops, and without major overhauls of less than 10,000 hours i ntervals Replacement of easily replaceable wearing parts such as cylinder suction and discharge air valves should not be required at intervals of less than 1000 hours. Replacement of piston rings should not be required at intervals of less than 1000 hours for the high pressure stages and of not less than 2000 hours for the first and second stages

"3.3. 1.1 The designer should take cognizance of the conditions under which equipment will be maintained and repaired on shipboard, and of the fact that the personnel responsible for maintenance and repair may not be seasoned mechanics. For example, parts should be so designed that it is impossible to replace them m a wrong position; and adjacent studs should be of the sa me length, or of large difference m lengths, so that a mistake will not easily occur; serrated surfaces should not be used for accurate Positioning unless specifically approved by the Bureau of Ships since areas should be avoided where strong wrenching would cause damage,

"3. .3. 1.2 In addition to the paramount concept of reliability, the design and construction of all compressors shall incorporate all the following requirements:

(a) Accessibility for repair
(b) satifactory operation when reclined as follows:
  (1) Up to 30 degrees in any direction (permanently inclined).
  (2) With the ship rolling up to 60 degrees from the vertical to either side, taking 8 seconds for a complete cycle (with direction of crankshaft in fore and aft position)
  {3) With the ship pitching 10 degrees up and down from the normal horizontal plane, taking 6 seconds for a complete cycle.
(c) Quiet operation,
(d) Minimum space and weight.

"3.3.1 3 To be in consonance with 3 3.1 2(d) the dry weight of compressors with 440 volt [illegible] current (a. c,) motors, less motor controllers, but including all other appurtenances, shall not exceed 4000 pounds. The corresponding weight with 230 v. direct current (d. c.) motors shall not exceed 5000 pounds

"3 3 1 4 Reliability analysis - In order to Achieve the desired reliability the contractor shall submit in analysis of those factors effecting reliability, as highlighted in 3 3, 1 The analysis shall or such that the composite reliability of the compressor, including essential appurtenances can be predicted in terms of mean-time-between-fadures (MTBF) MTBF is the time which equipment (parts, material) can be expected to operate satisfactorily before failure occurs

"3.3.1.4 1 The reliability analys sis shall include, but shall not be limited to the following

(a) List of those parts which experience and judgement show are subject to wear and service failures.
(b) Spec ific design features employed to attain the required service life of the parts Some suggested design features are choice of materials, compatability of materials, hardness, surface finishes, fits, clearances, and so forth. Include calculations when applicable
(c) Complete calculations of temperatures throughout the air path
(d) Cho ice and analysis of all bearings.
(e) Inter and after cooler design data
(f) How failure of one part affects other parts

"3.3. 1.4.2 The reliability analysis shall include preventive maintenance and servicing requirements necessary to the achievement of the predicted MTBF Any unusual steps or precautions necessary in carrying out maintenance and servicing requirements shall be pointed out.

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"3.3.1.5 The reliability analysis shall be submitted at the same time, or prior to, the submittal of class A and class D drawings for approval (see 3.7.7.2). Approval of the reliability analysis is requisite for approval of new class A and class D drawings, In the case of class A drawings approved prior to the requirement for reliability analysis, such reliability analysis shall be submitted to the Bureau of Ships for approval prior to using the class A drawings on any contrac t or order involving a bidding invitation dated subsequent to the date of this specification amendment."

Page 4, paragraph 3 3.4(c), after "Relative humidity": Delete "50 percent" and substitute "O 0 percent (dry air)".

Page 4, paragraph 3.3.5.1: Delete and substitute:

"3.3.5.1 The design of all complete air compressor units shall be such that they *are* capable of passing the type A, high impact (H. I.) shock tests for grade A, class III, equipment in accordance with MIL-S-901 A shock test with the compressor solidly mounted is considered proof of class III capability (see 4. 5.4). NAVSHIPS 250-660-30 may be used as a guide for design of shock resistant Naval equipment, "

Page 4, paragraph 3.3.6.2, last sentence: Delete and substitute:

'The compressor manufacturer will not be required to furnish any mounting members, motion limiting, or vibration buffering devices between the isolation mounts and the ship's structure, "

Page 4, paragraph 3.3.7, first sentence: Delete and substitute:

"Gage board temperature indicating and alarm instrument, dram valves and all features of an operational nature shall be located on or operable from one side, designated as the operating side of the compressor, so as to be accessible if the compressor is installed adjacent to a bulkhead. "

Page 5, paragraph 3.3.14,1, line 3: After "MIL-B-17931" insert "or FF-B- 185".

Page 5, paragraph 3.3.15.3, line 3. Between "bearings," and "gears" insert "wrist pin bearings, "

Page 6, paragraph 3.3.16, third sentence: Delete and substitute:

"All top and bottom bearing surfaces shall be machine finished, except bottom of subbases designed for center -of-gravity sound-isolation support "

Page 6, paragraph 3.3.17, third sentence" Insert a period after "bearings" and delete the remainder of the sentence,

Page 6, paragraph 3 3 19 1.3, line 3 Delete 'motor" and substitute "water"

Page 8, paragraph 3,3.20.5, first sentence: Insert a period after "minimum" and delete the remainder of the sentence.

Page 9, paragraph 3 3.25. Delete and substitute

"3,3.25 Piping -

"3.3.25 1 All piping and fittings shall conform to MIL-STD-438 or MIL-STD-777 except as modified herein. Where ultrasonic inspection requirements are invoked components to the following Bureau of Ships drawings shall be used: 810-1385941 to 810-1385946 inclusive, 810-1385948, 810-1385950 to 810-1325952 inclusive.

"3.3.25.2 Unions or flanges shall be used for compressor water connections to ex ternal shipboard piping sy stems The unions and flanges shall be in accordance with MIL - F- 1183 and MIL - F-.2094 2 respec:l".cl:, When space or weight savings ca n be effected, terminal c onnec tions *to* compressor component parts may hav e unions partially integrated with such parts, and may utilize the brazed or welded ring and male union nut type of construction with soft copper gaskets Tapered pipe thread connections shall not be used except for fittings such as plugs for core holes, or the dead ends of cross -drilled passages, which are intended to keep openings

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permanently closed, Terminals and joints of small tubing lines mounted on the compressor shall be straight thread flareless fittings conforming to MIL- F-21467 except that working pressures above 3000 psi will be permitted in the 1 4 inch outside diameter (o. d.) size only. The final discharge air connection to the ship s piping shall be a union connection m accordance with Drawing 810-1385884 with socket welding tail piece All piping shall be designed and arranged to permit maximum access ibility to compressor parts requiring periodic inspections or repair,

"3.3.25 3 All silver brazed piping connections shall conform to the far icati on and inspections requirements of NAVSHIPS 250-634-6 "

Page 9. paragraph 3 3.29, line 2. Delete "sulfuric"

Page 11. paragraph 3.5.2,1, line 7 Delete "(see 3 3 1.5 -l)' and substitute "(3 3 15 4)

Page 12. paragraph 3,6 1 Delete and substitute

"3.6.1 Onboard - When required by the contract or order (see 6 1), repair parts shall be furnished and processed m accordance with MIL-P- 15137, Each set of repair parts shall consist of the parts listed on the latest approved Allowance Parts List (APL) covering the particular equipment being furnished under the contract or order If no approved APL exists covering the particular equipment, then table 11 shall be used as the basis for provisioning forms required by MIL-P- 15137, and any other items considered necessary by either the bureau or agency concerned or the manufacturer to be essential in successful servicing of the equipment shall be included "

Page 16, paragraph 3 7.4.3.3, line 3 Delete "shall" and substitute "may"

Page 17, paragraph 4 1 Delete and substitute.

"4. 1 Responsibility for inspection - Unless otherwise specified in the contract or purchase or der, the supplier is responsible for the performance of all inspection requirements as spe cified herein Except as otherwise specified, the supplier may uti lize his own facil ities or any commercial laboratory acceptable to the Government The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements "

Page 19 paragraph 4 5 1 2, line 3 Delete "NAVSHIPS 250-648-2 and substitute "NAVSHIPS 250-

Page 19, paragraph 4 5.2.1, at the end of the first sentence change the period to a comma and add corrected to rated speed, receiver temperature zero humidity and 14 7 p s i a barometer last sentence Delete and substitute "The calibration of the test volume shall be at the [illegible] shall be made in the presence of the Government inspector

Page 15, paragraph 4.5 2.1, 1, line s 4 and 5 Dele te second and substitute third

Page 21 paragraph 4 5 3 c and - Delete 3 2 1 and substitute 3 2 1 2

Page 21 Add as paragraphs 4 5 3 10, 4.5 3 10 1 and 4 5 3 10 .2

"4 5 3 10 Castings, testing and repair - Castings having excess porosity sponginess or cracks will be cause for rejection Peening of such castings to repair defects will not be permitted Where minor porosity is evident as disclosed by weeping or sweating, castings may be repaired by impregnation in accordance with MIL-STD-276

"4 5 3 10 1 Fur water jacketed cylinder castings (less cylinder liners) where radiographic inspection is required as a prerequisite to impregnation but is not practical due to the design of the castings these castings shall be subjected to the proof test in 4 5 3 10 2 in lieu of radiographic inspection prior to machining

'4 5 3 10 2 First hydrostatically test the castings to a pressure of 1000 p s i g with the pressure applied slowly and maintained for 15 minutes Following this, the castings shall be inspected by dye penetrant

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test. Then if the castings pass both the dye penetrant test, without [illegible] indications, and the 1000 p.s.i.g. hydrostatic pressure test, they may be impregnated."

Page 21, paragraph 4.5.4, line 4. Delete "grade" and substitute "class" after last sentence, add "(see 3.3.5.1)"

Page 21, paragraph 4.5.4.4. Delete and substitute

"4.5.4.4 When the shock tests are on a medium-weight shock machine six blows shall be applied as described in MIL-S-901 with the mounting platform in a horizontal position and the blows applied under the approximate center-of-gravity of the assembled compressor unit. The equipment shall be tested running and at standstill on alternate blows. In addition to the six blows required above, two additional blows shall be applied with the base of the compressor unit inclined at 30 degrees from the horizontal and the blows applied under the approximate center-of-gravity. The compressor unit shall be mounted in such a way that the horizontal component of shock is transmitted across the axis in which the equipment is weakest. The blows of the inclined test shall correspond to groups II and III of test schedule for medium weight shock machine of MIL-S-901 and the compressors shall be running during both blows. Compressors need not be compressing air during shock tests."

Page 23, paragraph 6.1. Delete "(h) Whether repair parts are required. If required specify type (metal or wood)."

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